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A Detailed Study of Vulnerability Detection using Common Vulnerabilities and Exposures from NVD using Machine Learning and Deep Learning Models

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Abstract - In the recent era, usage and demand of software devices are growing rapidly. Due to cause this, chances of cyberattacks and the vulnerability is also growing at the rapid rate. Therefore, securing this type of sensitive data and detecting types and level of vulnerability is our prime responsibility. Common Vulnerabilities and Exposures (CVE) is one of the popular databases that widely used for predicting and tracking the software as well hardware vulnerability; The types and level of vulnerability can be identified by unique CVE-id number and CVSS Severity respectively. Vulnerability can be detected in the early stage by using machine learning and deep learning models. NVD provides the total number of cyber-attacks with their name that are responsible to damage data. In this paper list of ML and DL models or algorithms are discussed along with the performance of software and their future scope to improve the overall results and detect the risk factor at the initial stage.

Keywords - NVD, Common Vulnerabilities and Exposures, Machine Learning model, Deep Learning, Vulnerability, Cyber-attacks.

I. INTRODUCTION

In the digital world, more and more people depend upon the e-data due to the availability and ease of access. They rely to store their sensitive data on the online platform's sites or on their digital devices. Therefore, the demand of online platform and open software are increase rapidly. As a result, Security of such sites are become one of the challenging tasks for an individual. Maintain security and predicting vulnerability of the software as well as hardware and to secure physical machine and e-data from cyber-attacks is mandatory. There is a U.S based website

National Vulnerability Database (NVD) that store list of large databases that termed as software vulnerability as per the year [1] [2][3]. In each vulnerability database there is a unique vulnerability number that represents the type of cyber vulnerable attack that weak the system and it is maintained by National Institute of standards and technologies (NIST). It gives the list of vulnerabilities of hardware and software's [4].

There are various terms listed on the NVD database site that provides the dataset along with the list of vulnerability records. CVE is one of the import terms that stands for Common Vulnerabilities and Exposures. The dataset on CVE is handled by a team named MITER. The detail of vulnerability can be predicted by their unique identification number that is known as CVE-id [5][6]. Furthermore, Common Vulnerability Scoring System (CVSS) is a standard framework that decide the level of software's severity value and its weakness. It is in the form of numerical value that lie between 0.0 to 10.0 [7]. CVSS Score has critical range from 9.0 to 10.0; .1 to 3.9 shows the lowest severity ranging; 4.0 to 6.9 is medium CVSS severity and 7.0 to 8.0 is high risk factor [8][9]. CPE stands for Common Platform Enumeration. It is a naming standard used for the information system platforms, that includes the details of CPE entry, operating system, vendor name, product detail, language of product and version of the vulnerability [10]. The Common Weakness Enumeration (CWE) provide the vulnerability report that describe the security weakness of the software. It also measure the effectiveness of the software [11][12][13][14].

The vulnerability in a software, hardware and network is responsible for reducing the overall performance of device. Therefore, some of the modern techniques or model such as machine learning, deep learning technique is required for detecting the risk factor as in early stage.

A. *Some of the most popular machine learning and deep learning algorithms are listed below:*

- *Decision tree:* Decision tree perform its working on both types of datasets like categorical and continuous variable. The prediction of decision tree is like a tree style structure. Decision tree divide the data or dataset into sub groups based on the inputting data. This is non-parametric model whereas outlier has a less effect on the predicted model and in a same model multiple conditions can be implemented. ID3, C4.5, CART are most popular decision tree algorithms. This model is used for fraud detection model, risk assessment model [15].
- *XGBoost Machine Learning Algorithm:* This algorithm is form of ensemble machine learning method that uses the concept of gradient boosting algorithm. It always takes different weak learning models and generate a final strong one. It constructs the base model and decision tree. Its outcome is fetched as per the weight available of each node. It is one of the efficient and fast algorithms [5],[16].
- *SVM algorithm:* SVM can handle classification and regression related problems where SVM stands for support vector machine. It finds out the hyperplane that deals with boundary values and split the data into

different classes. SVM divide the whole data set in two hyperplane spectrum i.e. positive and negative hyperplanes. These two categories are decided as per the margin length [2].

- *Naive Bayes algorithm*: Naive Bayes algorithm is based on the bayes theorem that perform prediction on the basis of probability of data elements. This algorithm is applicable for the large data set. Spam filtration and sentimental analysis are the best example of naïve bayes algorithm [4].
- *Random forest algorithm*: Random Forest algorithm is groups of decision tree that predict the result on the trained tree only and return the result to that class included maximum values. It returns the results in form of single value [5].
- *K-means:* K-means is type of unsupervised machine learning model that deals with all clustering related problems. In this algorithm k represents the number of datasets; where centroid is calculated and compare it with each data point and make grouping according. It follows the iterative that means continue until value of centroid's value remain same. Customer behaviour, image segmentation and compression can be performed with this algorithm [10].
- *Artificial Neural Network:* Artificial neural network is generally used to predict result by the previous recorded data using the trained machine. Classification, clustering, recognising patterns, prediction result can be performed with the help of ANN model. ANN is applied if huge data set are available as data set elements. Artificial network implemented on the numerous hidden layers that are connected with input layer and after calculate this we get the final results [1], [7], [17] .
- *Convolutional Neural Network*: CNN is one of the oldest neural networks that are implemented by the concept of neurophysiological. CNN is working on 2-dimentional matrix or grid type structure. CNN involve video and image for recognise pattern. Detection of pattern of image in CNN model is performed with the concept of pixel [18], [19], [20].

II. Statistical Analysis of top vulnerabilities

National Vulnerability Database (NVD) has a list of cyber risk and it can be identified by their CVE-ID. This is storing latest data as per the type, date and by their unique number CVE id listed below:

A. *Vulnerability from the last five years*

CVE Vulnerability in context of cybersecurity that is responsible for weak the performance of system Vulnerability can be depended upon various factor like lack of security, lack of awareness, error in software and so on. NVD store total number of vulnerabilities from last five years as shown in graph.

B. *Types of vulnerability from last five year*

Figure 2 shows the type of top cyber-attack that make our system weak from the last five years. Data is calculated from the NVD site and as per the previous result XSS is most critical attack in 2020 an XXE is weak cyber security attack.

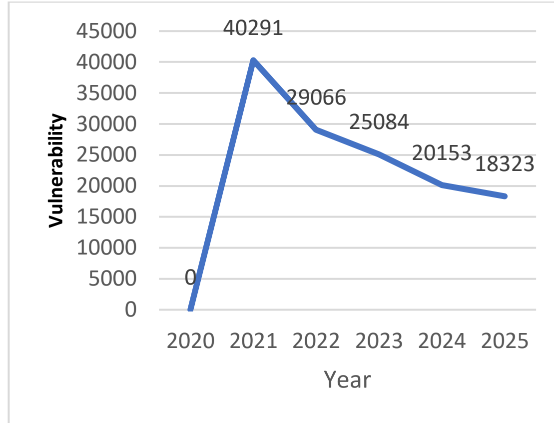

Fig. 1. Total number of CVE from 2020 to Jan 2025 form NVD Database [21]

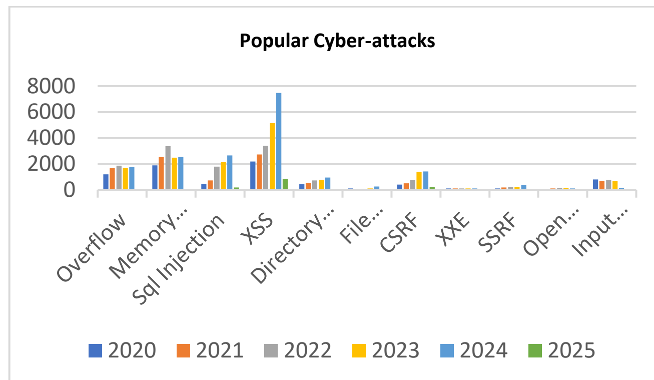

Fig. 2. Most popular cyber-attacks from 2020-2025 as per NVD database [22]

C. *Top Cyber-attacks as per NVD dataset*

CVE_ID of cyber-attacks are given along with their date of publication in 2024. The CVSS severity is also available in the tabular form. The Severity of the cyber-attack can be identified by its numerical value; whose range lie between 0-10. Top 20 cyber-attacks along with details are given below:

TABLE I. Top 20 vulnerability cyber-attacks in 2024 as per NVD results [23]

CVE-ID	***Year of published***	***Description***	***CVSS Severity***
CVE-2024-47574	November 13, 2024	Unauthorized access to device and steal the data of victim.	7.8 High
CVE-2024-32117	November 12, 2024	Path Traversal vulnerability (CWE-22).	4.9 Medium
CVE-2024-32116	November 12, 2024	Multiple relative path traversal vulnerabilities (CWE-23).	6.0 Medium
CVE-2024-31496	November 12, 2024	Stack-based buffer overflow vulnerability	6.9 Medium
CVE-2024-23666	November 12, 2024	It is a client-side vulnerability that allows attackers to gain unauthorized access to Fortinet products.	8.8 High

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CVE-2023-44255	November 12, 2024	The attack allows an attacker to read logs events of another using HTTPs requests.	4.1 Medium
CVE-2024-33506	October 8, 2024	Remote authenticated attacker vulnerability.	4.3 Medium
CVE-2024-45327	September 11, 2024	Brute Force Attack	7.5 High
CVE-2023-23775	June 11, 2024	SQL Injection	8.8 High
CVE-2024-23664	June 3, 2024	Open Redirect' vulnerability (CWE-601)	6.1 Medium
CVE-2024-31493	June 3, 2024	An improper removal of sensitive information before storage or transfer vulnerability (CWE-212)	6.5 Medium
CVE-2024-31488	May 14, 2024	Cross-Site Scripting (XSS) vulnerability (CWE-79)	9.0 Critical
CVE-2024-51540	December 26, 2024	This is an arithmetic overflow vulnerability (CWE-190) found in Elastic Cloud Storage (ECS).	6.5 Medium
CVE-2024-52534	December 25, 2024	Authentication Bypass by Capture-replay vulnerability (CWE-294) identified in Dell's Elastic Cloud Storage (ECS)	5.4 Medium
CVE-2024-11364	December 19, 2024	Uninitialized Variable vulnerability (CWE-457) code execution vulnerability	7.3 high
CVE-2024-11157	December 19, 2024	An Out-of-Bounds Write vulnerability (CWE-787)	7.3 high
CVE-2024-8748	December 02, 2024	Buffer overflow vulnerability that causes a temporary denial of service (DoS) attack.	7.5 High
CVE-2024-9197	December 02, 2024	A post-authentication buffer overflow vulnerability	4.9 Medium
CVE-2024-9200	December 02, 2024	It is post-authentication command injection vulnerability in certain Zyxel devices.	7.2 High
CVE-2024-1645	March 11, 2024	This vulnerable attack is dealing with mobile attack for WordPress plugin that gives the permission to attacker to access without taking authority from the user.	4.3 Medium

III. Review of Literature:

In this section, the motive of this research is to predict automatic risk and vulnerability detection and prevention at the early stage. Classification of vulnerability can be identified with their published year can be download from the NVD website; and this type of early detection can be possible with the help of machine learning and deep learning models. The detail research of implemented models along with the future scope are listed below:

T.B. Fraunk et al. [1] This paper focuses on automatic detection of unstructured questions by generating a question answer system and provide answers accordingly using deep learning model such as BERT, GPT-2, T5. BERT for generating automating question answer detection system for reducing cyber security attacks and as per the results BERT model provide overall results in form of exact match and F1 that is approximately 65% and 80% respectively. A.N. Kia at al. [2] paper emphasis on to detect vulnerability of dataset by creating Prediction model CyRiPred along with wikiTopic sub-module is used as a text extraction algorithm.

This model developed model is used to improve efficiency and to identify future risk based on time series extraction. The model followed by M. A et al. [3] is GPT-2 Model along with deep learning approach is used for the text extraction model and predicting high severity vulnerability. Furthermore, this model can be extended to other natural language processing model for the better prediction. In paper [4] empirical study, comparative study is performed on CVE database for predicting vulnerability using ML algorithm due to the Machine Learning classification technique is suitable technique for automatic vulnerability prediction and in paper [5] comparative study is performed between two machine learning classification models Random Forest Classifier and Gradient Boost Classifier. As per the experiments Gradient Boost Classifier give consistent and better results instead of Random Forest classifier.

The X. Wang et al. [6] performs experiment-based study to identifies security patches in open-source software (OSS) and ML approach is applied for reducing vulnerability on OSS. The author is working on ten types of vulnerabilities such as buffer overflow, SQL injection, cryptographic techniques and so on by implementing ML algorithms using C++ language for reducing inconsistency in the dataset. In research article [7] Deep Learning technique i.e neural network is implemented using C programming for automatic vulnerability detection. Two experiments granularity refinement and leveraging the intermediate code-based representations are used for the immediate vulnerability detectors. Furthermore, in research article M. Marali et al. [10] both clustering and classification machine learning technique is implemented for detecting vulnerability and classification in software security. Dataset is taken from NVD database repository and as per the result performed using machine learning technique Linear SVM gives the best accuracy whereas bernoulli naïve bayes algorithm gives the worse accuracy. F. A. Alshaya et al. [11] published his research work in 2023 and this research article use an automatic tool name vulnerability report tagget (VrT) for identification cybersecurity labels and to assign the vulnerability text via generating vulnerability report. VrT tool is implemented using Python and it aids to improve accuracy and identifying cyber security text.

W. Zheng et al. [12] emphasis on empirical study to performed on traditional machine learning and deep learning techniques to detect vulnerability of computer software by taking dataset on NVD and SARD website. In machine learning Random Forest, Adaboost and support vector machine implement and in Deep Learning CNN, GRU and BLSTM used. Author is working with three types of vulnerability with CWE id is: 119, 399 and 664. AS per the result shows that BLSTM performs better result than machine learning model.

In paper [13] research study is performed on the popular deep learning model Implemented Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) method using python and the purpose of this study is to identifying vulnerability in computer software using NVD dataset and in future, experiment can be performed on large and complex dataset; and some study is required for improve the performance of imbalanced dataset. Moreover, the latest research is performed by M. Babu et al. [17] in 2024 and this paper emphasis the detect vulnerability issues such as buffer overflow, SQL injection, Cryptographic issues with the help of XGBoost ML algorithm. XGBoost ML model helps to improve the results in terms of precision, recall, F1 and accuracy and its overall result is more than 90%. In future, real-time system can be generated for tracking vulnerability, improve prioritization, and taking appropriate decision. Several deep learning models like BERT, XLNet and GPT-2 NER models are used in 2024 by W. Hu et al. [18] and GUI model is also implemented for visualization. out of all these models BERT gives better result. However, this empirical study is performed on small dataset only; so, in future, more result is required for generating fast and efficient result. L. Sandholm et al. [19] is used for automatic risk is evaluated in early stage by using dataset of NVD website and for calculating result binary classification model is applicable. This is because Classification method is used for high precision and better result. In future, clustering can also be implemented to perform same experiments. In paper [20], the weakness stored in vulnerability articles are discussed. MITRE View-1003 is taken from NVD and binary classifier model named Cross-encoder models is applied. Cross-encoder models is used for predicting similarity between CVE description and CWE information and calculating CWE score those aids to use for predicting weakness of vulnerability. GPT models can be used for the future for performing the same experiments.

IV. Conclusion

In conclusion, detection of vulnerability and risk assessment at an initial stage is essential; otherwise maintaining security of e-devices and software's will become one of the challenging tasks. NVD is one of the popular websites that has availability of vulnerability list along with their id along as well as publishing year of particular attacks. The main purpose to choosing CVE dataset is availability of latest dataset. In this review paper, various ML and DL algorithms are discussed for predicting risk; and as per the result top vulnerability from the last five years are cross-site scripting (XSS) cyber-attack in 2020 to 2025 and XXE has minimum CVSS Severity. In future, more empirical study can be performed for generating real time system for vulnerability detection and other type of complex dataset can be used for the better and fast results.

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